

FÍSICA Y QUÍMICA

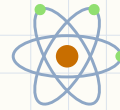
2.º ESO

Class notes

$$d = \frac{m}{V}$$

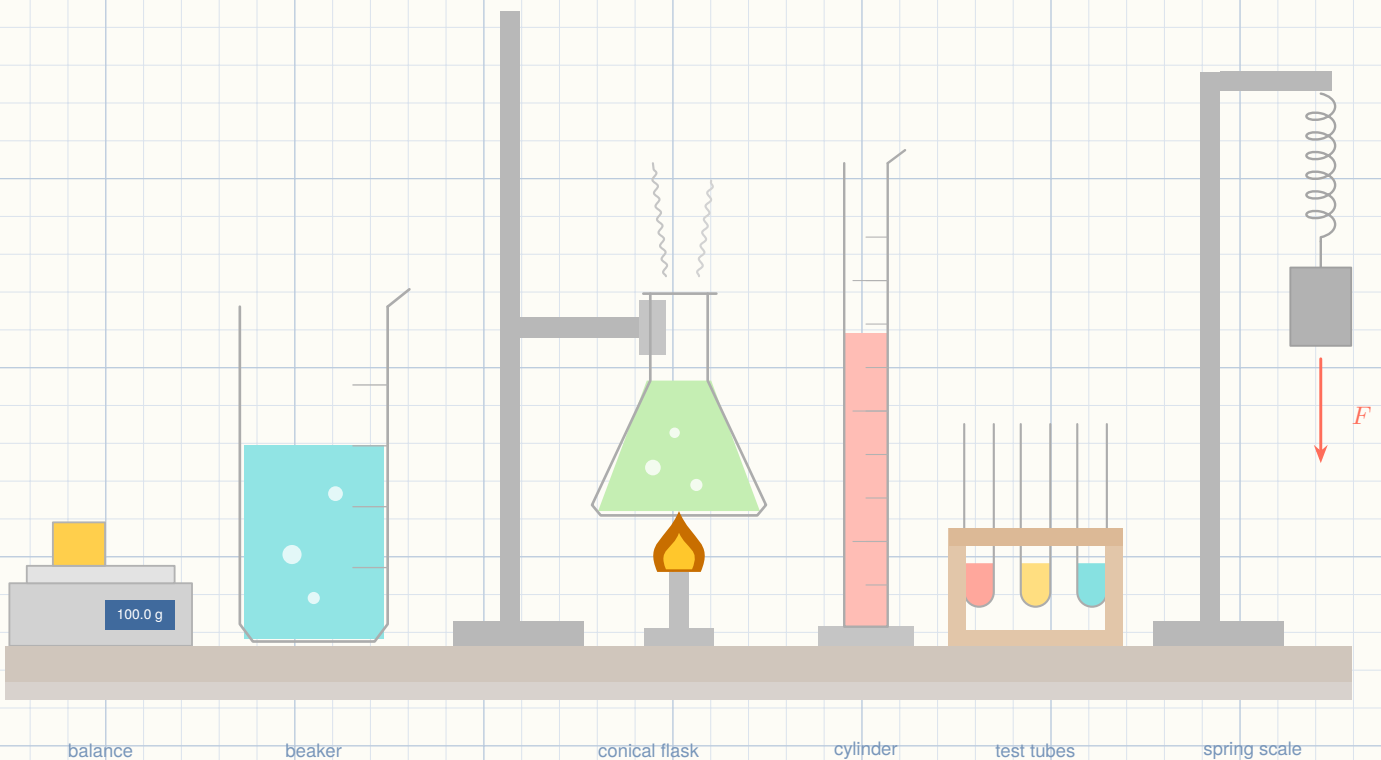
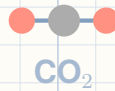


$$F = m \cdot a$$



$$v = \frac{d}{t}$$

$$P \cdot V = k$$



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UNIT 1

Matter and measurement

“ The spacecraft that got lost because of its units

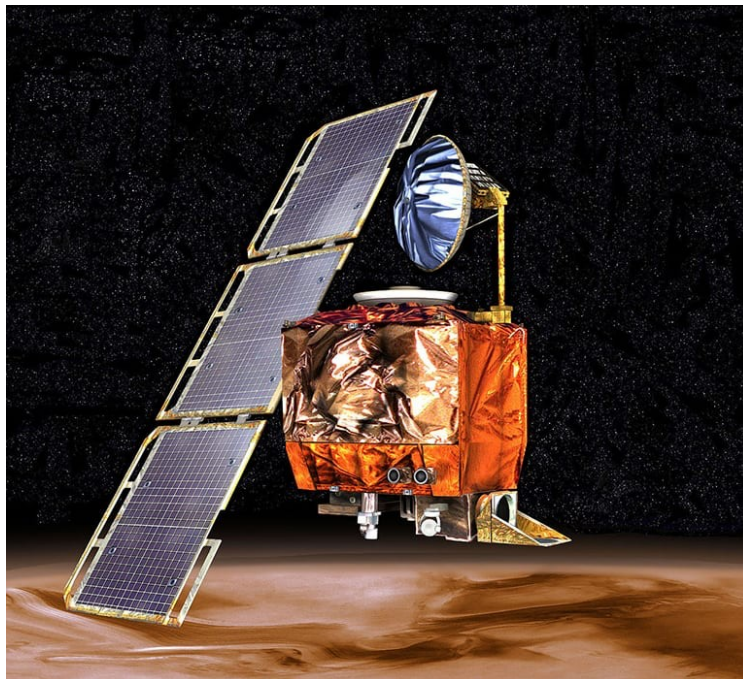
In 1999, NASA wanted to study the weather on Mars. They built a spacecraft called Mars Climate Orbiter. Its job was to look at the atmosphere, the clouds, and the surface of the red planet.

The project was very expensive. It cost about 327 million dollars, and hundreds of scientists and engineers worked for years to make it possible. Everything was ready, and the spacecraft began its long trip to Mars.

But when the spacecraft arrived, something very strange happened. Instead of flying safely around Mars, it went too close to the atmosphere. In a few seconds, it was lost forever.

*What was the problem? It was not the technology, it was not the cameras, and it was not the rocket. The problem was **the units**. One team used pounds of force (English system), while the other team used newtons (International System). They were speaking two different “languages of measurement”, and because of this, the calculations were wrong and the spacecraft was destroyed.*

*This story teaches us something very important: if scientists, engineers and students do not use the same units, mistakes can be very big. That is why the **International System of Units (SI)** is so important. It gives us one clear and common language, so people around the world can work together without errors.*



1.1 Magnitudes and measurement

Magnitude

A **magnitude** is any property of an object or a phenomenon that we can **measure** and express with a **number** and a **unit**.

To measure means to compare a quantity with another quantity of the same magnitude, which we take as a reference and call the **unit**.

Examples of magnitudes:

- The length of a table: 2 m (unit: metre, symbol: m).
- The mass of a book: 1.5 kg (unit: kilogram, symbol: kg).
- The time of a race: 12.3 s (unit: second, symbol: s).

Examples of things that are *not* magnitudes (we cannot measure them with a number and a unit): beauty, happiness, kindness, sense of humour.

1.2 The International System of Units (SI)

In the past, every country (sometimes every town!) had different units of measurement. Nowadays, the units are standardized: they show no variations and they are clearly defined. This helps commercial activity and scientific communication.

This standardized way of measuring is called the **International System of Units (SI)**. It uses **seven base units**:

Base magnitude	Unit	Symbol
time	second	s
length	metre	m
mass	kilogram	kg
electric current	ampere	A
temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

The rest of the units are obtained from these base units, and they are called **derived units**. For example, square metres (m^2) are used to measure surface area, and cubic metres (m^3) are used to measure volume.

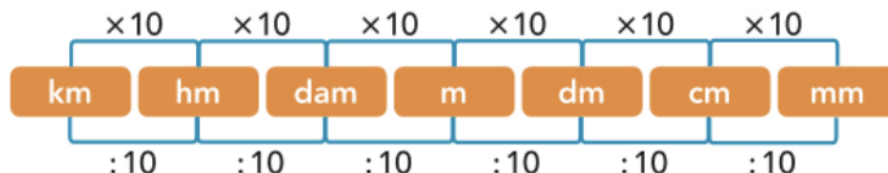
! Be careful!

If a magnitude is not in the table above, it is **not** a base magnitude. For example: is speed a base magnitude? Speed is measured in m/s. Is the unit m/s in the table? No, so speed is a **derived** magnitude (it comes from length divided by time).

1.3 Length, surface, volume and mass

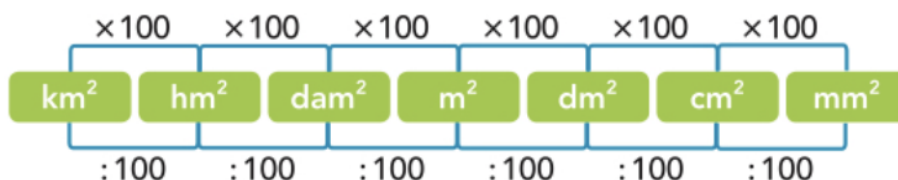
Length

Length is the magnitude that expresses the distance between two points. The SI unit of length is the **metre (m)**, but we often use multiples (km, hm, dam) or submultiples (dm, cm, mm). They are based on the decimal system: each step multiplies or divides by 10.



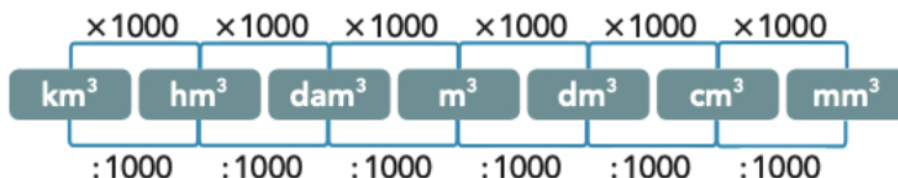
Surface area

A **surface** is the outside part of an object or a shape. The SI unit of surface area is the **square metre (m²)**. To change between multiples and submultiples, we multiply or divide by **100** in each step (because a square has two dimensions).

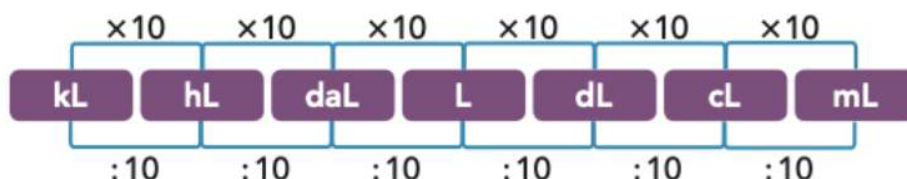


Volume

Volume is the space that a body occupies. The SI unit of volume is the **cubic metre (m³)**. To change between multiples and submultiples, we multiply or divide by **1000** in each step (a cube has three dimensions).



A very common unit to measure volume is the **litre (L)**. We change between its multiples and submultiples by 10, in the same way as with metres.



It is very useful to know the equivalence between litres and cubic units:

Equivalent units

$$1 \text{ m}^3 = 1 \text{ kL}$$

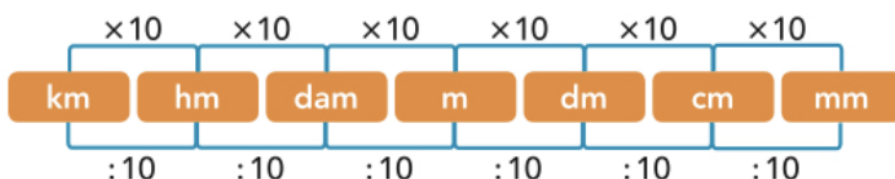
$$1 \text{ dm}^3 = 1 \text{ L}$$

$$10 \text{ cm}^3 = 1 \text{ cL}$$

$$1 \text{ cm}^3 = 1 \text{ mL}$$

Mass

Mass is the amount of matter in a body. The SI unit of mass is the **kilogram (kg)**. Its multiples and submultiples also work in steps of 10.



! Be careful!

Do not confuse **mass** with **weight**. Mass is the amount of matter (measured in kg). Weight is a force (measured in newtons). We will study this difference in Unit 7.

1.4 Conversion factors

Conversion factors are a mathematical method used to change units. They are very useful for difficult conversions, where we could make mistakes if we tried to do them with mental calculation.

A conversion factor is a fraction that is equal to 1, because the numerator and the denominator express the same quantity. We multiply our quantity by this fraction, and the old unit cancels out.

Conversions step by step

$$1300 \text{ g} \rightarrow \text{kg} : 1300 \text{ g} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} = 1.3 \text{ kg}$$

$$3 \text{ m} \rightarrow \text{mm} : 3 \text{ m} \cdot \frac{1000 \text{ mm}}{1 \text{ m}} = 3000 \text{ mm}$$

$$5 \text{ m}^2 \rightarrow \text{cm}^2 : 5 \text{ m}^2 \cdot \frac{10\,000 \text{ cm}^2}{1 \text{ m}^2} = 50\,000 \text{ cm}^2$$

$$2 \text{ m}^3 \rightarrow \text{L} : 2 \text{ m}^3 \cdot \frac{1000 \text{ L}}{1 \text{ m}^3} = 2000 \text{ L}$$

$$72 \text{ km/h} \rightarrow \text{m/s} : 72 \frac{\text{km}}{\text{h}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ h}}{3600 \text{ s}} = 20 \frac{\text{m}}{\text{s}}$$

Notice the last example: when the unit is a fraction (km/h), we need **two** conversion factors, one for each unit.

1.5 Scientific notation

Scientific notation is a way to write very big or very small numbers easily. It uses powers of 10. It is useful in science to simplify calculations and to avoid mistakes with many zeros.

Rule

Move the decimal point so that there is **one non-zero digit** before the decimal point. Then multiply by 10 raised to the number of places you moved the decimal point. If you moved it to the **left**, the exponent is **positive**; if you moved it to the **right**, the exponent is **negative**.

Scientific notation

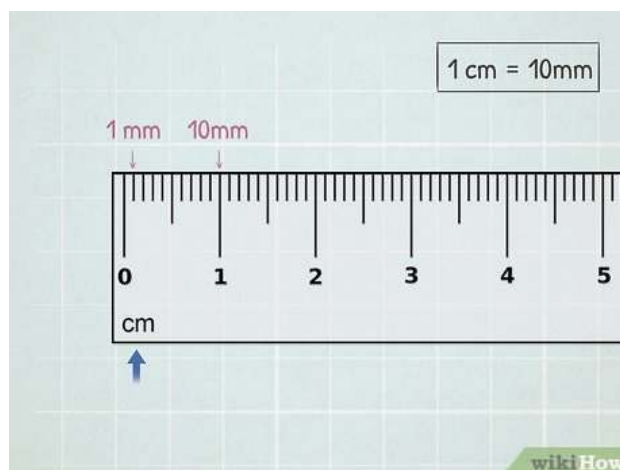
- $1000 = 1 \times 10^3$. We move the decimal 3 places left: $1000 \rightarrow 1.000$, so we multiply by 10^3 .
- $0.001 = 1 \times 10^{-3}$. We move the decimal 3 places right: $0.001 \rightarrow 1.0$, so we multiply by 10^{-3} .
- $72\,000 = 7.2 \times 10^4$. We move the decimal 4 places left.
- $0.00056 = 5.6 \times 10^{-4}$. We move the decimal 4 places right.

1.6 Sensitivity of a measuring device

Sensitivity

The **sensitivity** of a measuring device is the **smallest change** in the quantity that the device can detect. It is the value of the smallest division of its scale.

Let's understand it with an example.

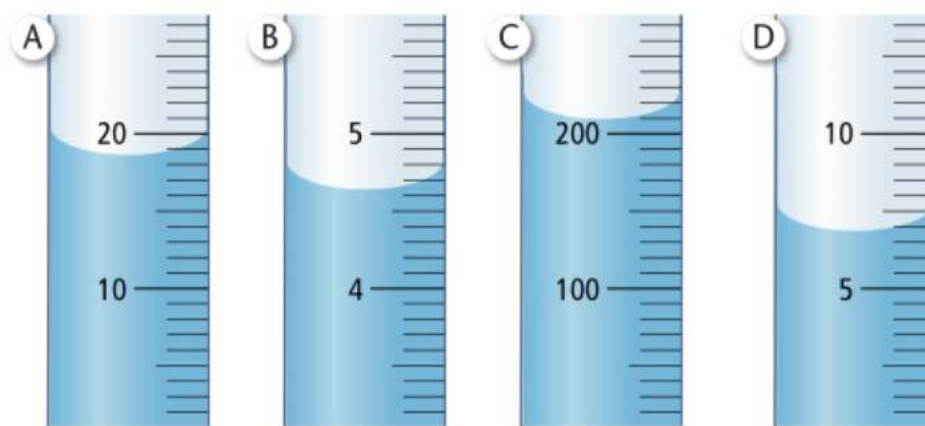


Look at the ruler in the image. The numbers 1, 2, 3... indicate centimetres. Each centimetre is divided into 10 parts, so each part is one millimetre (1 mm). This is the smallest change that the ruler can detect, so the sensitivity of the ruler is **1 mm**.

What is the length L of the black rectangle? $L = 2.4\text{ cm}$. **We cannot add more decimals**, because we cannot measure more precisely than the sensitivity of the device.

And if the end of an object seems to be between 1.1 and 1.2 cm? Can we write 1.15 cm? **The answer is no.** The sensitivity of the ruler is 0.1 cm, so we cannot add more decimals — we would be making up the measurement! We choose the closest mark, for example 1.1 cm.

Let's check if you have understood it. What is the sensitivity of each cylinder? What is the volume of water in each cylinder? The cylinders are graduated in millilitres (mL).



1.7 Rounding

Rounding means making a number simpler while keeping it close to the original value. We do this because our instruments cannot measure with infinite precision.

Rounding rule

Look at the first digit you want to remove:

- If it is **5 or more**, we round **up**.
- If it is **4 or less**, we round **down**.



 Rounding with a thermometer

A thermometer has a sensitivity of $0.1\text{ }^{\circ}\text{C}$. We measure the temperature three times: $22.1\text{ }^{\circ}\text{C}$, $22.4\text{ }^{\circ}\text{C}$ and $22.6\text{ }^{\circ}\text{C}$.

Step 1: calculate the average.

$$\frac{22.1 + 22.4 + 22.6}{3} = \frac{67.1}{3} = 22.3666 \dots\text{ }^{\circ}\text{C}$$

Step 2: round to the sensitivity of the instrument. The thermometer only shows one decimal place. The second decimal is 6, so we round up.

Final answer: the average temperature is $22.4\text{ }^{\circ}\text{C}$.

1.8 Lab equipment

In the laboratory we use special equipment. You must learn the names of the most common pieces and what they are used for.

CONTAINERS



flask



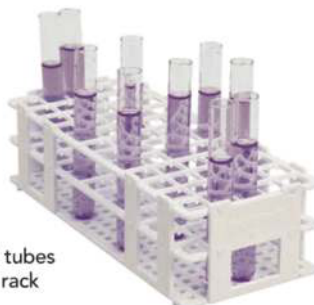
beaker



Erlenmeyer or
conical flask



crystallising dish



test tubes
in a rack



separatory
funnel



Petri dishes

HEATING EQUIPMENT**EQUIPMENT FOR DIFFERENT USES**

Why are they made of glass?

Most laboratory glassware is made of **borosilicate glass**, a type of glass known for its:

- **Thermal resistance:** it can resist high temperatures and sudden temperature changes without breaking.
- **Chemical durability:** it does not react with most chemicals, so the experiments stay pure.
- **Transparency:** it allows scientists to observe reactions clearly.

The purpose of their shapes

- **Narrow neck** (volumetric flasks, Erlenmeyer flasks): allows greater precision when measuring liquids, because the narrow opening helps control the meniscus and reduces error.
- **Wide neck** (beakers, test tubes): makes pouring and mixing easy, but gives less accuracy in volume measurements.

Why is a temperature indicated?

Many glassware items, especially volumetric flasks, include a temperature marking (for example, “20 °C”). This is because volume changes with temperature: the marked volume is correct only at that specific temperature.

1.9 Lab safety



1. The first thing to do is to **locate the safety equipment** (eyewash station, extinguisher, etc.) and the personal protection items (lab coats, goggles, gloves, etc.). Find out what they do and how to use them.
2. **Listen carefully** to your teacher's instructions. You must be clear about the purpose of the activity, the procedure and the rules about the use of materials.
3. **Prepare the materials** you will need. Make sure everything is clean and well ordered. This way, everything will be nearby and you will not need to move around the lab.
4. **Follow the protocol** and do not do any unauthorized experiments.
5. **Never eat or drink** in the lab, to prevent accidental poisoning.
6. After the activity, **clean your desk** and the equipment you have used. Follow your teacher's instructions about waste disposal.

7. **Wash your hands** with soap and water before leaving the lab.

1.10 Why chemistry is a science, but palm reading is not

Some people believe that the lines on your hand can tell your future or your personality. This is called **palm reading**. It can be fun, but it is **not a science**. Chemistry *is* a real science. What is the difference? Let's see.

What is science?

Science is a way to learn about the world using **evidence** and **experiments**. In science we can measure things, repeat the experiments, and check whether an idea is right or wrong.

A **pseudoscience** looks like science, but it does not follow its rules. Palm reading is a pseudoscience. In the table, chemistry answers “yes” to every question, while palm reading answers “no”.

Question	Chemistry (a science)	Palm reading (a pseudoscience)
Can we measure it?	Yes, with instruments (mass, volume, temperature...).	No; the lines mean different things to different readers.
Can it make clear predictions?	Yes: vinegar and baking soda always make gas.	No; the predictions are vague (“something good will happen”).
Can we repeat it with the same result?	Yes, anywhere in the world.	No; two readers give different answers.
Can it be proven wrong?	Yes; one experiment against it is enough.	No; it never gives a clear yes or no.

The 5 steps of the scientific method

1. **Observe** something and ask a question.
2. Make a **hypothesis**: a possible answer.
3. Do an **experiment** to test the hypothesis.
4. **Analyse** the results (measure and compare).
5. Write a **conclusion**: was the hypothesis right or wrong?

Hypothesis, law and theory

These three words do not mean the same thing in science:

- A **hypothesis** is a possible answer that we still have to test.
- A **law** describes what always happens, often with a formula (for example, the gas

laws in Unit 2).

- A **theory** explains *why* it happens and is supported by a lot of evidence (for example, the kinetic-molecular theory).

In science, a **theory** is not “just an idea”: it is a strong, well-tested explanation.

Glossary — 10 key terms

1. **Magnitude** — a property we can measure and express with a number and a unit.
2. **To measure** — to compare a quantity with a unit taken as a reference.
3. **Unit** — the reference amount used to express a measurement, for example the metre.
4. **Base unit** — one of the seven main SI units, not obtained from any other unit.
5. **Derived unit** — a unit made by combining base units, for example m/s or m².
6. **International System (SI)** — the standard set of units used in science all over the world.
7. **Sensitivity** — the smallest change a measuring device can detect; its smallest division.
8. **Rounding** — making a number simpler while keeping it close to the original value.
9. **Scientific method** — the way scientists test ideas: observe, make a hypothesis, do an experiment, and reach a conclusion.
10. **Hypothesis** — a possible answer to a question that scientists still have to test with an experiment.

1.11 Exercises

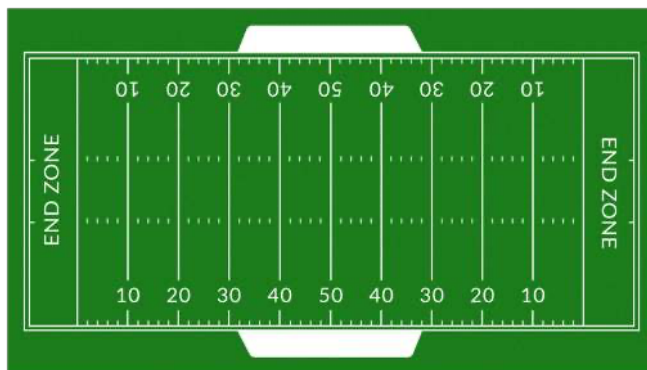
Exercise 1. What is the SI? What are the advantages of an international system of units?

Exercise 2. Look at these properties. In your notebook, write down the ones that are magnitudes: taste – speed – kindness – sadness – volume – temperature – time – intelligence – sense of humour.

Exercise 3. How are base units and derived units different? Give an example of each.

Exercise 4. Calculate:

- a) There is an American football field. The distance between the lines is 10 yards. Calculate the distance in metres from one end zone to the other. (1 yard = 0.914 m.)
- b) There is a running race in Albacete. The race is 3 miles long. How many kilometres is this? (1 mile = 1.609 km.)



Exercise 5. Find the mistakes. There is a distance of 25 kilometres between two towns. The following statements are wrong in some way. Find the errors and correct them in your notebook.

- Length is a base unit.
- 25 Km is the correct way to abbreviate this distance.
- The towns are 25 kms apart.
- The unit km is derived from kilograms and metres.

Exercise 6. Convert the following quantities using conversion factors:

- | | |
|-----------------|------------------|
| a) 52.5 cm to m | e) 1250 g to kg |
| b) 3.2 m to cm | f) 2.3 kg to g |
| c) 5.75 km to m | g) 750.5 mg to g |
| d) 2300 m to km | h) 12.8 m to cm |

Exercise 7. Convert the following quantities using conversion factors:

- | | |
|---|------------------------------|
| a) 2.5 m^2 to cm^2 | e) 2.3 L to m^3 |
| b) 7500 cm^2 to m^2 | f) 0.0045 m^3 to L |
| c) 0.35 km^2 to m^2 | g) 0.002 m^3 to L |
| d) $1\,250\,000 \text{ m}^2$ to km^2 | h) 850 mL to L |

Exercise 8. Convert the following quantities using conversion factors:

- | | |
|---|--|
| a) 36 m/s to km/h | e) 0.0045 kg/L to g/mL |
| b) 72 km/h to m/s | f) 2.3 g/mL to kg/L |
| c) 0.85 kg/m^2 to g/cm^2 | g) $25 \text{ m}^3/\text{h}$ to L/s |
| d) 1.2 g/cm^2 to kg/m^2 | h) 0.36 L/s to m^3/h |

Exercise 9. Fill the gaps applying scientific notation:

3×10^4			68 300
	370 000		0,000505
3×10^{-4}			375 000 000
	630	$3,45 \times 10^1$	
	67 800		46,35
2×10^{-2}			463,5
	170		435
$5,6 \times 10^1$			460
	30 100	$5,674 \times 10^{-1}$	
	41 000		0,4635
$5,093 \times 10^6$			0,004635
	0,0079	$4,5 \times 10^{-6}$	
	9,5		0,017
$9,98 \times 10^{-3}$			9808

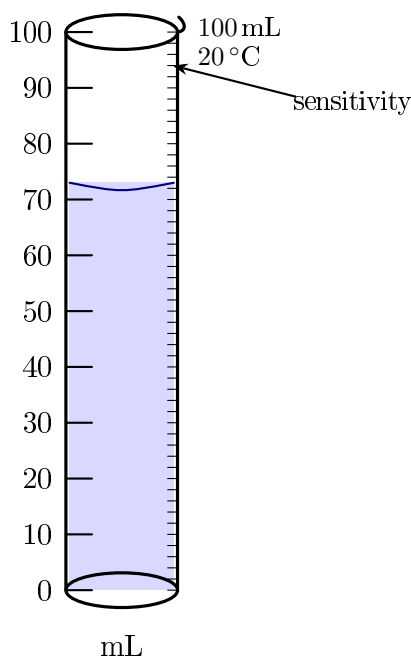
Exercise 10. Round the following numbers to the required digit of each column:

Number	To the hundredth	To the unit
6.541		
0.745		
9.604		
23.067		

Exercise 11. For each situation, decide the appropriate rounding based on the context or the sensitivity of the measurement. Then solve.

- You measured temperatures in a lab three times: 21.8°C , 22.3°C , 22.0°C . Calculate the average and round it.
- You need to split €47.85 between 4 friends. Round the amount each person should get.
- You timed a race and got 12.43 s, 12.57 s, and 12.49 s. Find the average and round it.
- A cake recipe asks for 250.7 g of sugar, but your scale only measures in whole grams. Round the amount.

Exercise 12. A graduated cylinder is a piece of laboratory equipment used to measure the volume of liquids. This cylinder measures in millilitres.



- What is the maximum volume this graduated cylinder can measure?
- Look at the number of divisions between two numbered marks on the scale (for example, between 90 and 100) and determine the cylinder's sensitivity.
- What is the volume of the liquid in the cylinder?
- Why is the volume of a graduated cylinder specified at a certain temperature (20 °C)?

Exercise 13. All together. For each measurement below: first **round** it to the sensitivity shown, then write the rounded value in **scientific notation**, and finally **convert** it to the unit asked, using conversion factors.

- | | |
|--|--|
| a) 1348.6 g (sensitivity 1 g) → kg. | d) 0.73914 L (sensitivity 0.001 L) → mL. |
| b) 0.004736 m (sensitivity 0.0001 m) → mm. | e) 126.7 km/h (sensitivity 1 km/h) → m/s. |
| c) 2.4568 km (sensitivity 0.001 km) → m. | f) 3.482 g/cm ³ (sensitivity 0.01 g/cm ³) → kg/m ³ . |

Exercise 14. The smartphone factory. A factory makes 2500 phones per day, and each phone has a mass of 168 g.

- What is the total mass of one day's phones, in kilograms?
- Write that mass in scientific notation.
- The phones are packed in boxes of 20. How many full boxes are made in one day?
- Each phone comes with a charging cable 0.85 m long. What total length of cable is used in one day? Give the answer in km.

Exercise 15. Three instruments. In the lab you take three measurements: a balance shows 45.3 g (it can read to 0.1 g), a ruler shows 12.7 cm (it can read to 1 mm), and a stopwatch shows 8.24 s (it can read to 0.01 s).

- a) Write the sensitivity of each instrument.
- b) A classmate writes the mass as 45.312 g. Explain why this is wrong.
- c) Convert the length to metres and to millimetres.
- d) Write the three measurements in scientific notation.

UNIT 2

States of matter and gas laws

“ Five experiments to think about

The balloon in hot and cold water. Place a balloon over the mouth of an empty glass bottle. If you put the bottle in hot water, the balloon slowly inflates. When you move it into cold water, the balloon deflates again. Why? The air inside expands when heated because its particles move faster and spread out, pushing the balloon upward. Cooling makes the particles slow down, so the balloon shrinks.

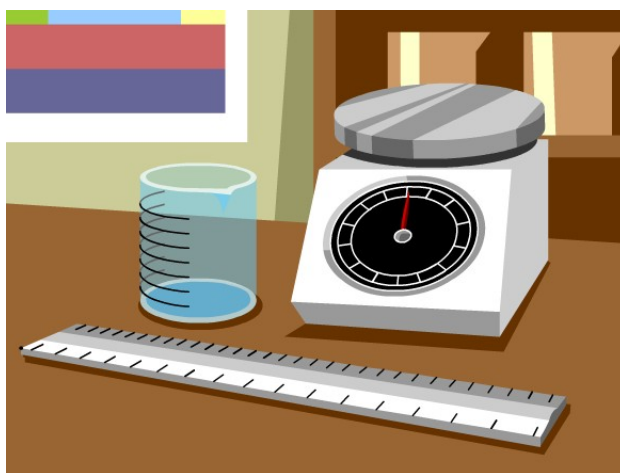
The plastic bottle on an airplane. Take a closed plastic bottle with you on a flight. As the plane climbs, the air pressure outside decreases, so the bottle looks swollen. During descent, the outside pressure increases and squeezes the bottle inward.

The egg in the bottle. Light a small piece of paper inside a glass bottle, then quickly place a peeled boiled egg on the opening. As the flame goes out, the pressure inside drops. The higher external pressure pushes the egg inside the bottle.

The ice cube in warm water. Drop an ice cube into a glass of warm water: the ice melts faster than in cold water. Heat from the liquid flows into the solid, breaking the bonds that hold the water particles in place.

The metal ball and the ring. A metal ball just fits through a metal ring. When the ball is heated, it expands and can no longer pass through the ring. After cooling, it contracts and fits again.

All these experiments show that matter changes with temperature and pressure. Science discovered a simple truth: **matter is made of particles**. If we understand how these particles move and behave on a very small scale, then we can understand how matter behaves on a very large scale.



Matter

Matter is everything that has **mass** and occupies a **volume**.

In this unit we will learn to **measure** matter, study how it **changes state**, and discover how **gases** behave.

PART 1 Measuring and density

Properties of matter

The properties of matter are classified as:

- **Extensive (general) properties:** they are common to all types of matter and only depend on the **amount** of substance. *Mass* and *volume* are extensive properties.
- **Intensive (specific) properties:** they are specific to a particular type of matter and do **not** depend on the amount. They help to differentiate one substance from another. Some intensive properties are colour, odour, hardness and *density*.

We will focus on the following properties:

Mass

Mass is the amount of matter that a body has. It is measured in kg in the SI. We measure it with a scale (a balance).

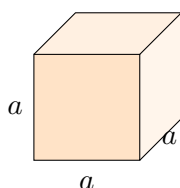


Volume

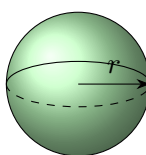
Volume is the amount of space that a body occupies. It is measured in m^3 in the SI, although we will often find it in litres (L) or millilitres (mL).

How do we measure volume? It depends on the state of aggregation:

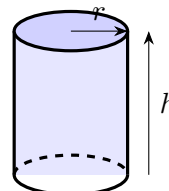
- **Liquids:** we measure their volume using the correct laboratory equipment, for example a measuring cylinder.
- **Solids with a regular shape:** we use maths formulas. We will study the cube, the sphere and the cylinder:



$$V = a^3$$

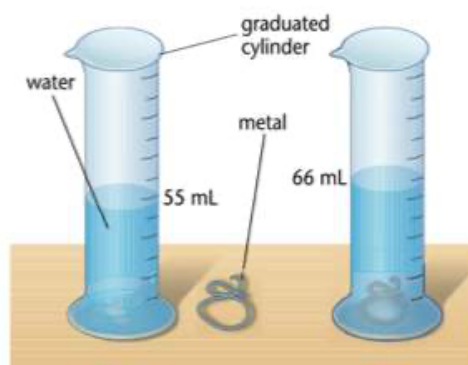


$$V = \frac{4}{3}\pi r^3$$

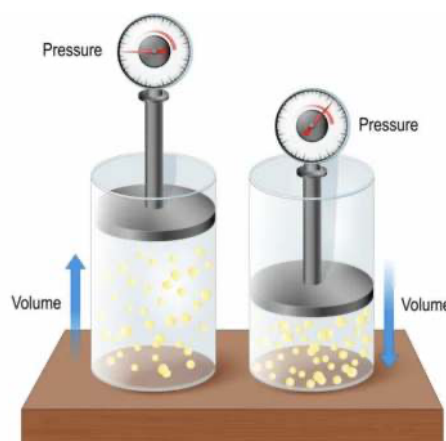


$$V = \pi r^2 h$$

- **Solids with an irregular shape** (like a rock): we put the solid into a liquid and look at how much the liquid level goes up. This method is called **water displacement**. If the solid is fully underwater, the rise in volume is the volume of the solid.



- **Gases:** gases fill all the volume of their container, so the volume of a gas is the same as the volume of its container. But this volume changes a lot when pressure or temperature change, so we study gases separately at the end of this unit.



Density

Density

Density relates the mass of a body to its volume:

$$d = \frac{m}{V}$$

It is measured in kg/m^3 in the SI, although we will often use g/cm^3 . ($1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$.)

Every pure substance has its **own** density. This way, knowing the density of a pure substance, we can guess what substance it is — density works like a “fingerprint”.

Density is related to **floating**: less dense materials float on more dense ones. The bigger the difference in densities, the better they float. That is why it is so easy to float in the Dead Sea: its water has a lot of dissolved salt, so it is much denser than normal sea water.



Density of some substances (at 20 °C)	
Substance	Density (g/cm ³)
gold	19.3
lead	11.3
iron	7.8
zinc	7.1
aluminium	2.7
pinewood	0.5
beech wood	0.7
water	1
seawater	1.025
olive oil	0.91
air (at sea level)	0.0012

Using density

A piece of metal has a mass of 54 g and a volume of 20 cm³. What is its density? Could it be aluminium ($d = 2.7 \text{ g/cm}^3$)?

Formula: $d = \frac{m}{V}$

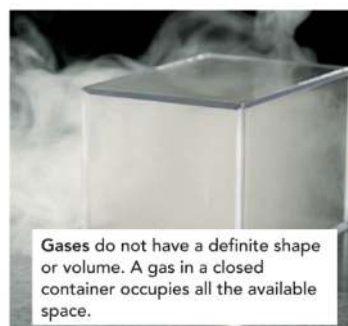
Substitution: $d = \frac{54 \text{ g}}{20 \text{ cm}^3}$

Result: $d = 2.7 \text{ g/cm}^3$. Yes, it could be aluminium, because the density is the same.

PART 2 Changes of state and the kinetic-molecular theory

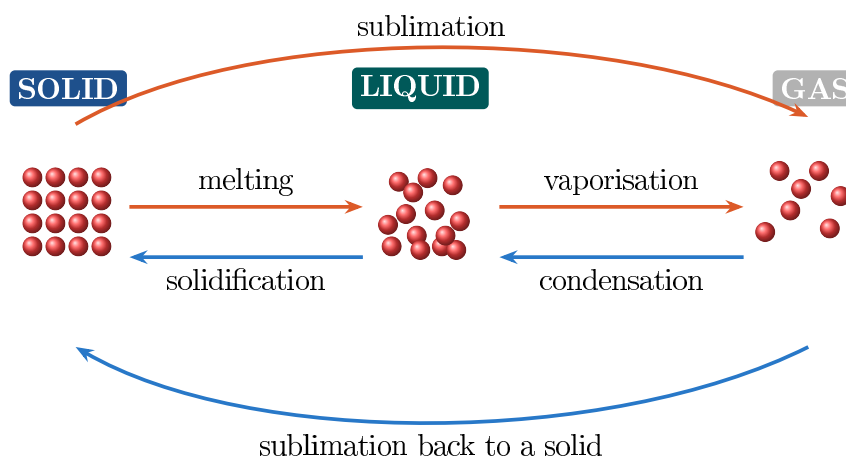
The states of matter

States of aggregation: matter can be found in three different states: **solid**, **liquid** or **gas**.



	Solid	Liquid	Gas
Shape	fixed	takes container's shape	takes container's shape
Volume	fixed	fixed	fills all the container
Can it flow?	no	yes	yes
Can we compress it?	no	almost no	yes, easily

Change of state: matter can change its physical state. We can find water, for example, as a solid (ice), a liquid or a gas (water vapour). Each change of state has a name:



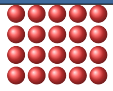
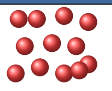
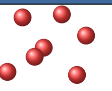
The kinetic-molecular theory

Richard Feynman, a famous scientist and Nobel Prize winner, was once asked: “If you had to explain all science in one sentence, what would it be?” He answered: “*All matter is made of atoms.*”

This idea is the basis of the **kinetic-molecular theory**. This theory explains how matter works at a microscopic level. If we understand the microscopic world, we can explain things we see in the macroscopic world, like temperature, pressure or expansion.

Kinetic-molecular theory: four main ideas

1. All substances are made up of **particles**.
2. The particles are in **continuous motion**. In solids, they vibrate without changing place.
3. **Temperature** is a measure of the speed of the particles. The higher the temperature, the faster they move.
4. Particles attract each other with **cohesive forces**, which are very strong in solids, moderate in liquids and very weak in gases.

Solid	Liquid	Gas
		
The particles in solids are strongly bonded in rigid structures. The only movement they can make is a slight vibration.	The particles in liquids are not as strongly bonded as those in solids, so they can move freely around each other without forming stable structures.	The particles in gases are independent of each other and move freely in all directions, bumping into the walls of the container that contains the gas.

★ Curious fact

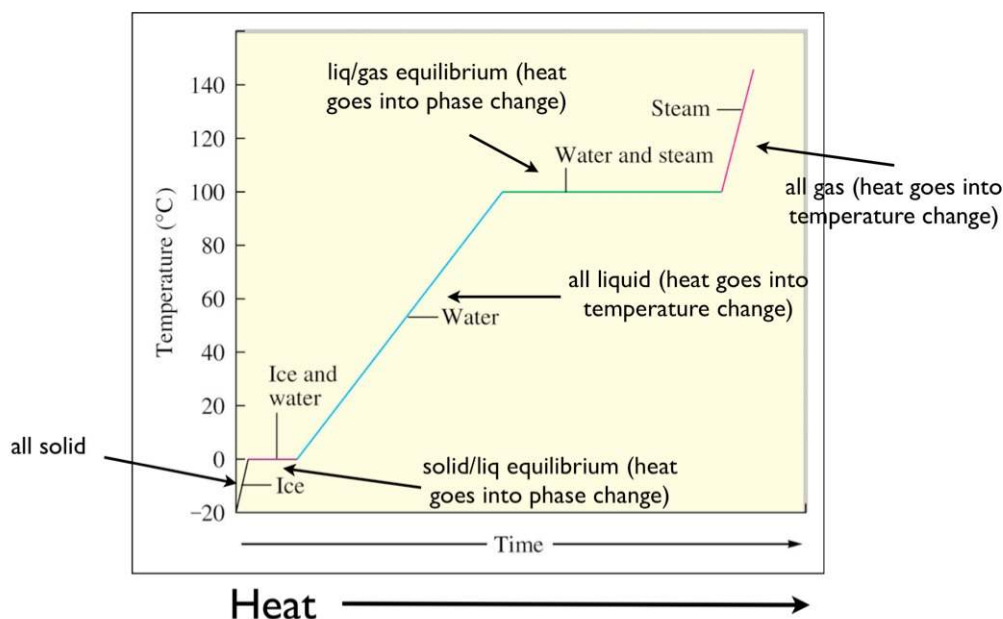
Is there a minimum temperature? Yes! If we cool matter down, the particles move more and more slowly. At -273°C the particles would stop completely, so it is impossible to go lower. This temperature is called **absolute zero**, and it is the zero of the **kelvin scale**: $0\text{ K} = -273^{\circ}\text{C}$. To change from Celsius to kelvin: $T(\text{K}) = T(^{\circ}\text{C}) + 273$.

How does the theory explain changes of state?

If we want a substance to change its state, we need to make its particles be very far apart and move fast in all directions (like in a gas), or very close and only vibrating (like in a solid). **We normally do this by changing the temperature.**

If we heat a solid, it becomes a liquid. If we keep heating, it becomes a gas. The same happens in the opposite direction if we cool it down. We can also change the state by changing the pressure, like when we make liquefied gas (the gas inside a gas bottle).

We can understand how changes of state happen by studying the **heating curve of water**:



Let's start with an ice cube at -20°C and heat it:

1. All the energy we give to the ice makes its particles vibrate faster. The temperature rises until it reaches 0°C .
2. At 0°C the ice starts to melt. While the ice is melting, **all** the energy is used to break the forces that hold the particles together. The particles do not move faster, so **the temperature stays the same** (this is the first flat part of the graph).
3. When all the ice is liquid and we keep heating, the temperature of the liquid rises again.
4. At 100°C the water starts to boil. While it is boiling, all the energy is used to break the forces between the liquid particles, and again the temperature stays constant (second flat part).
5. When all the liquid has become gas, the temperature of the gas rises again.

Melting point and boiling point

The **melting point** is the temperature at which a substance changes from solid to liquid (or from liquid to solid).

The **boiling point** is the temperature at which a substance changes from liquid to gas (or from gas to liquid).

These points are like a border: they are the same for melting and freezing, and the same for boiling and condensing. Each pure substance has its own melting and boiling points.

Melting points and boiling points		
Substance	Melting point ($^{\circ}\text{C}$)	Boiling point ($^{\circ}\text{C}$)
oxygen	-218	-183
butane	-138	0
water	0	100
ethanol	-114	78
mercury	-39	357
sulfur	115	445
iron	1538	2862

* temperatures measured at sea level

The values in the table are measured at sea level. Melting and boiling points depend a lot on pressure. For example, water boils at 100°C at sea level, but on the top of Mount Everest it boils at about 80°C .

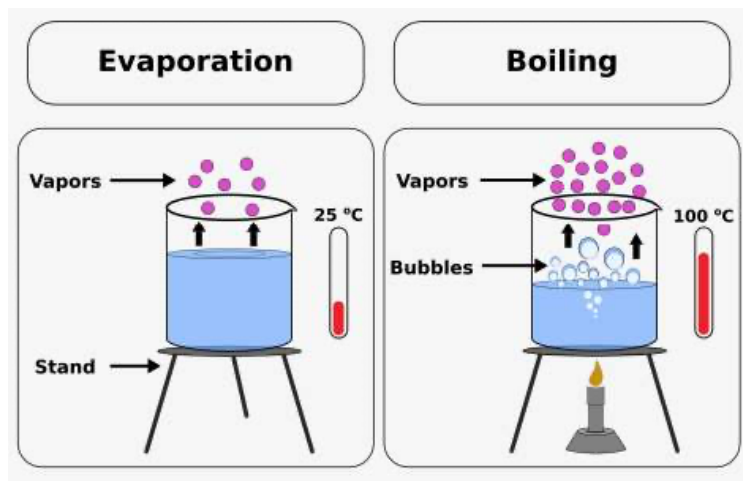
Vaporization, evaporation and boiling

It is important to know the difference between these three words:

- **Vaporization** is the general name for a liquid turning into a gas. It can happen in two ways: boiling and evaporation.
- **Boiling** happens when a liquid turns into gas **at its boiling point**. During boiling, all the liquid in the container is turning into gas (you can see bubbles everywhere).

- **Evaporation** happens when a liquid turns into gas **below the boiling point**. Only the liquid at the **surface** changes into gas. For example, water from the sea evaporates and later forms clouds.

So, boiling is like when you boil water in a pot, and evaporation is like water slowly disappearing from a glass of water.



Everyday changes of state

Fragmentation of rocks

One of the causes of rock fragmentation in cold areas happens when rainwater enters cracks in the rocks and freezes when the temperature falls below 0 °C. Most substances decrease in volume when they change from a liquid to a solid, but **water is an exception**: the volume of ice is about 9% bigger than the volume of liquid water. As its volume increases, the frozen water pushes out from the crack until the rock gradually breaks. This phenomenon is called **frost weathering**.



Refreezing

If you take some snow and press it into the shape of a ball, some of the snow melts due to the pressure. When you release the pressure, the liquid water freezes again and this ice cements the snowball together. The same idea explains why a very cold metal object from the freezer sticks to your hand: your body heat melts some ice, and then it refreezes between your skin and the object.

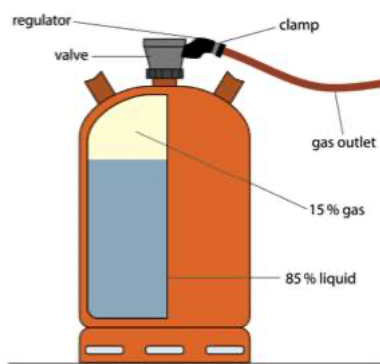
Drying clothes

Clothes dry due to the **evaporation** of the water we use to wash them. The water evaporates because of the sun and the wind. The clothes dry faster if they are spread out, because evaporation is a **surface** phenomenon. Modern tumble dryers push a flow of hot air across the wet clothes; the wet air is then cooled quickly so that the water vapour **condenses** in a separate tank.



Camping gas (butane)

The gas we use for cooking, such as **butane**, is stored as a **liquid** inside the bottle. By pressing the gas very hard (high pressure), it **condenses** into a liquid, so a lot of gas fits in a small bottle. When we open the valve, the pressure drops and the liquid turns back into gas. This is an everyday application of the **condensation** of gases.



Geysers

Geysers are produced by water that fills vertical cracks in active volcanic terrain. The water at the bottom heats up past its normal boiling point, but it cannot boil because the weight of the water above increases the pressure. The water overheats until it finally wins, and steam and hot water spray out of the geyser's vent.

What do we breathe out when it is cold?

We breathe out carbon dioxide and water vapour. If the weather is very cold, a small cloud comes out of your mouth that looks like smoke, but it is actually water vapour that **condenses** instantly into millions of tiny drops of liquid water.



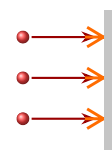
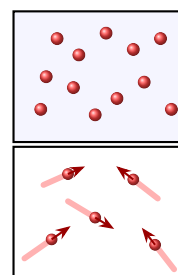
PART 3 The gas laws

In this part we study the three properties that define a gas: **volume**, **temperature** and **pressure**.

Volume (V): the volume of a gas is the same as the volume of the container that holds it. The same amount of gas can occupy very different volumes.

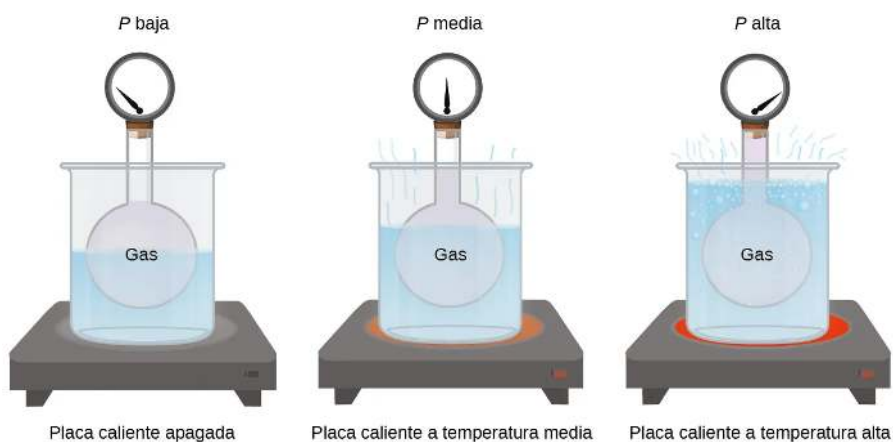
Temperature (T): it is a measure of how fast the particles move. The higher the temperature, the faster the particles move.

Pressure (P): pressure is a measure of how many times the gas particles hit the walls of the container, and how strong these hits are. Pressure is measured in **pascals (Pa)** in the SI. However, it is very common to use **atmospheres (atm)** or **millimetres of mercury (mmHg)**. You should know that $1 \text{ atm} = 760 \text{ mmHg}$.

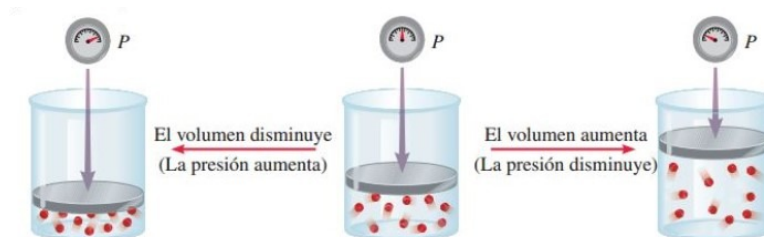


Pressure is closely related to volume and temperature:

- **When the temperature is high**, the particles move faster, so there are more and stronger hits on the walls (and vice versa).



- **When we increase the volume**, the gas particles have more space, so there are fewer hits on the walls (and vice versa).



In summary

- The number of particles inside the container never changes: the containers are always closed.
- When the volume increases, the pressure decreases. They are **inversely proportional**: if the volume doubles, the pressure becomes half.
- When the temperature increases, the pressure increases. They are **directly proportional**: if the temperature doubles, the pressure doubles.

! Be careful!

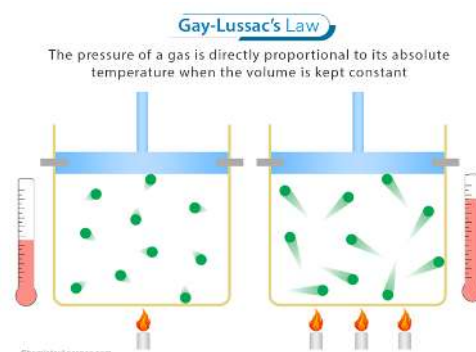
In all gas law calculations, the temperature must be in **kelvin (K)**, never in degrees Celsius. Remember: $T(K) = T(^{\circ}C) + 273$.

Law of constant volume (Gay-Lussac's law)

- **Volume**: constant.
- **Temperature**: varies.
- **Pressure**: varies.

Example to imagine: gas in a rigid metal container, like an oxygen tank or a pressure cooker from which no steam escapes.

Formula: $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

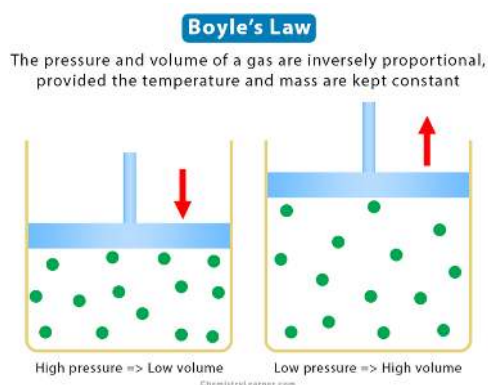


Law of constant temperature (Boyle's law)

- **Volume**: varies.
- **Temperature**: constant.
- **Pressure**: varies.

Example to imagine: the air inside a syringe when you push the plunger, or the air inside a closed plastic bottle when you squeeze it.

Formula: $P_1 V_1 = P_2 V_2$

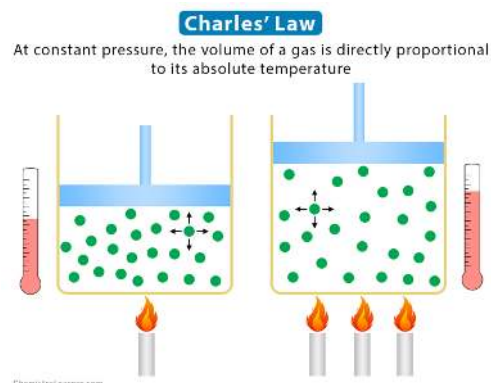


Law of constant pressure (Charles' law)

- **Volume:** varies.
- **Temperature:** varies.
- **Pressure:** constant.

Example to imagine: heating or cooling a balloon with air inside.

Formula: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$



Units: pressure in atmospheres (atm), temperature in kelvin (K) and volume in litres (L).

Gas law problem, step by step

A closed rigid container holds a gas at 27°C and 1.5 atm. We heat it to 127°C. What is the new pressure?

Step 1: identify the law. The container is rigid, so the volume is constant → Gay-Lussac's law.

Step 2: change the temperatures to kelvin. $T_1 = 27 + 273 = 300 \text{ K}$; $T_2 = 127 + 273 = 400 \text{ K}$.

Step 3: formula, substitution, result.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{1.5}{300} = \frac{P_2}{400} \Rightarrow P_2 = \frac{1.5 \cdot 400}{300} = 2 \text{ atm}$$

It makes sense: the temperature went up, so the pressure went up too.

Glossary — 10 key terms

1. **Mass** — the amount of matter a body has; it is measured in kilograms (kg).
2. **Volume** — the amount of space a body occupies; measured in m³ or litres.
3. **Density** — the mass of a body divided by its volume; each substance has its own.
4. **State of aggregation** — the physical form of matter: solid, liquid or gas.
5. **Temperature** — a measure of how fast the particles of a body move.
6. **Melting point** — the temperature at which a substance changes between solid and liquid.
7. **Boiling point** — the temperature at which a substance changes between liquid and gas.
8. **Vaporization** — a liquid turning into gas; it can happen by boiling or by evaporation.
9. **Evaporation** — vaporization only at the surface of a liquid, below its boiling point.
10. **Pressure** — how often and how hard gas particles hit the walls of their container.

2.1 Exercises

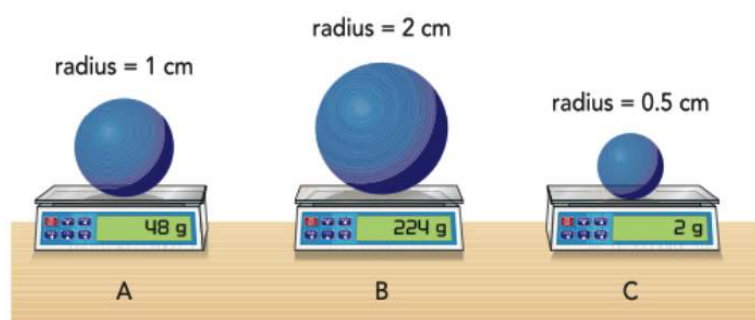
Part 1 — Measuring and density

Exercise 1. Write in your notebook whether the following statements are true or false, briefly explaining your answer:

- a) The density of a litre of water is greater than that of a drop of water.
- b) The density of a kilogram of water is less than that of a gram of lead.
- c) The density of a kilogram of water is the same as that of a kilogram of lead.
- d) A kilogram of lead has the same mass as a kilogram of water.
- e) A kilogram of lead occupies the same volume as a kilogram of water.

Exercise 2. We have weighed three spheres of different sizes and materials: A, B and C. In your notebook, do the following:

- a) Order them from greatest to least mass.
- b) Order them from largest to smallest volume.
- c) Calculate the density of each one and order them from highest to lowest density.



Exercise 3. Hardness is a property of solids that indicates their resistance to scratching. Is hardness an extensive or an intensive property? Justify your answer.

Exercise 4. A metal cube with a 2 cm edge has a mass of 22 g. What is the density of the metal? Can we know with absolute certainty, only from its density, what kind of metal it is?

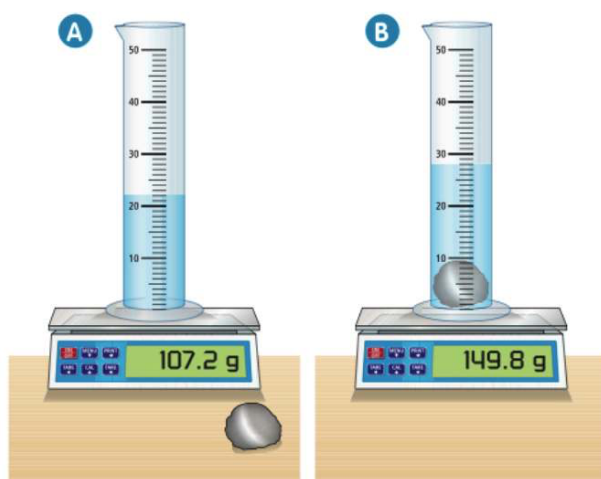
Exercise 5. A metal cylinder has a mass of 600 g, a diameter of 4 cm, and a height of 10 cm. First, draw the cylinder and label its dimensions on the drawing. Then calculate its density in g/cm^3 and, using the internet, compare the density you obtain with different metals to try to guess what metal it might be.

Exercise 6. Use the table of densities to complete this table in your notebook:

mass (g)	volume (cm ³)	density (g/cm ³)	substance
25	25		
100			gold
	50	0.91	
	4000		air

	Solid	Liquid	Gas
Distance between particles			
Movement of particles			
Strength of the force of attraction between particles			
Characteristics of volume and shape			

Exercise 7. A group of students found a small, irregularly shaped piece of metal and decided to determine its density to identify the metal. They filled a 100 cm³ graduated cylinder with some water and weighed it. Then, they put the piece of metal in the cylinder and weighed it again. They calculated the density of the metal from the data and identified it by looking at the density table. Which metal was it?



Metal	Density (g/cm ³)
aluminium	2.7
copper	8.9
zinc	7.1
iron	7.8
lead	11.3

Exercise 8. To cool a glass of water, we add some ice cubes. After a while, the ice cubes melt and the temperature of the water decreases. Explain whether or not you agree with these statements:

- The mass of water at the beginning and at the end is the same.
- The volume of water at the beginning and at the end is the same.
- The density of water at the beginning and at the end is the same.

Exercise 9. Explain whether the following statements are correct or not. (Look up the densities you need in the table below.)

- A litre of petrol weighs more than a litre of water.
- One cm³ of alcohol weighs less than 1 cm³ of water.
- One hundred grams of iron weigh less than 40 g of water.
- One cm³ is equal to one mL.
- 2.5 litres of milk weigh more than 3 litres of alcohol.

- f) Fifty grams of aluminium occupy more volume than 50 mL of oil.
- g) Oil is denser than water.
- h) Five kilograms of oil fit in a 5-litre container.
- i) One mL of water weighs one gram.

TABLA DE DATOS de DENSIDADES en g/cm³

Aluminio 2,7	Plomo 11,4	Mercurio 13,6	Hierro 7,9	Agua 1	Aceite 0,9	Alcohol 0,79
Oro 19,32	Níquel 8,9	Gasolina 0,68	Bromo 3,12	Acetona 0,8	Plata 10,5	Leche 1,03

Part 2 — Changes of state and the kinetic-molecular theory

Exercise 10. Solids have a fixed and permanent shape, like a rock. But let's look at an hourglass made with beach sand. We can see that the sand takes the shape of the container, like a liquid. Is sand a solid or a liquid? How can you explain this?



Exercise 11. Answer the questions about everyday physical changes:

- a) Explain why the bathroom mirror steams up when you take a hot shower.
- b) A lot of outdoor bars and restaurants have misting systems (humidificadores) that they use on very hot days. Explain the scientific reasons for their usefulness.
- c) Explain the benefit you get from sweating. What could happen to you if you do a physically demanding sport without drinking water for a long time?

Exercise 12. We use our senses to detect the matter around us, but we cannot see or touch everything. Give an example of matter that you can perceive by touch, but not by sight.

Exercise 13. What is the state of matter of snow? What is the state of honey? Justify your answers.

Exercise 14. How are evaporation and boiling similar? How are they different? Give examples of both processes. Will the boiling point be the same as the condensation point? Why?

Exercise 15. Explain the following phenomena:

- a) Why does a bottle or can of drink break if we leave it in the freezer for too long?
- b) Why does a bottle of pharmaceutical alcohol not break under the same circumstances?
- c) Why does frost form on cold nights?
- d) Why does popcorn explode in the microwave?
- e) In what state is the water found in clouds?
- f) Why do we blow on a spoonful of hot soup before eating it?
- g) Why does mist form when we breathe on a mirror?

Exercise 16. When water boils, bubbles form at the bottom that rise to the surface and burst. What is inside a bubble of boiling water? Represent this phenomenon in your notebook and indicate the different states of water.

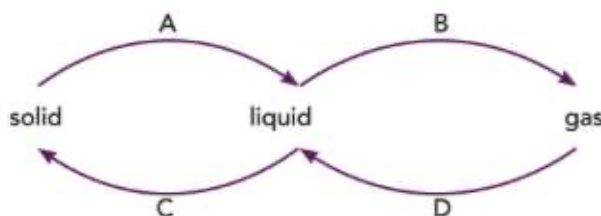
Exercise 17. The Earth's core is solid, even though the temperature there is so high that any matter should be in a gas state. How can you explain this? (Hint: think about what else, apart from temperature, can change the state of a substance.)

Exercise 18. The melting and boiling points of ethanol are -114°C and 78°C , respectively. Indicate the state of ethanol at the following temperatures: -200°C , 0°C , 70°C , and 95°C .

Exercise 19. Say whether these statements are true or false. Substances expand because...

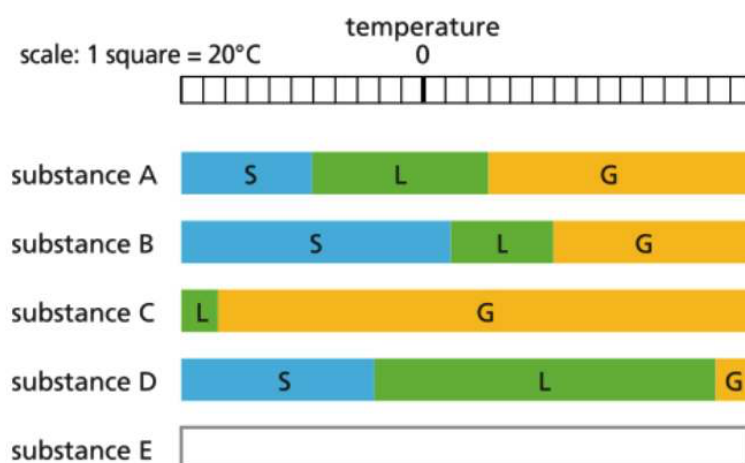
- a) ... the size of the particles increases.
- b) ... air is introduced.
- c) ... the particles separate as their movement increases.
- d) ... the density of the substance increases.
- e) ... the force of attraction between the particles decreases.

Exercise 20. Consider the changes of state represented by the letters A, B, C and D:



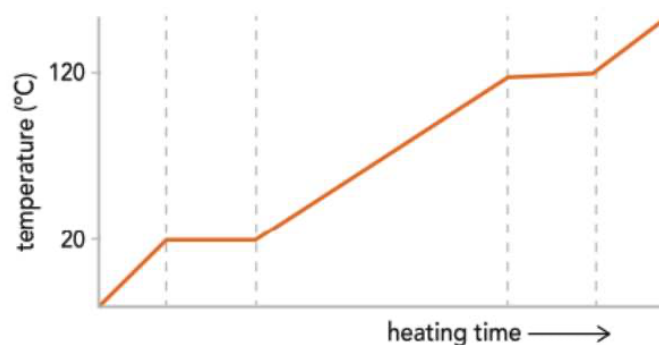
- a) What are the names of the changes of state represented by the letters A, B, C and D?
- b) Which of these changes are achieved by heating? Which are achieved by cooling?

Exercise 21. The diagram below represents the physical state of different substances according to temperature:



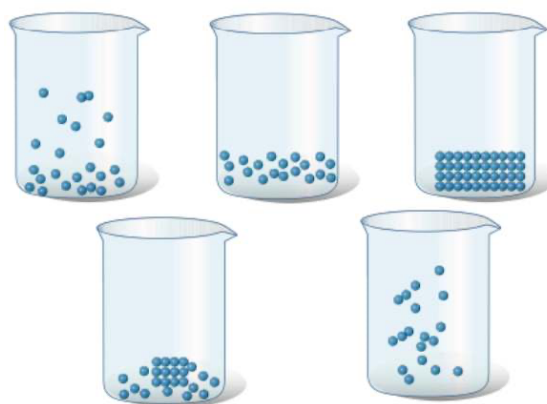
Analyse the information in the diagram and answer the questions:

- a) Which substance does the following heating chart represent?



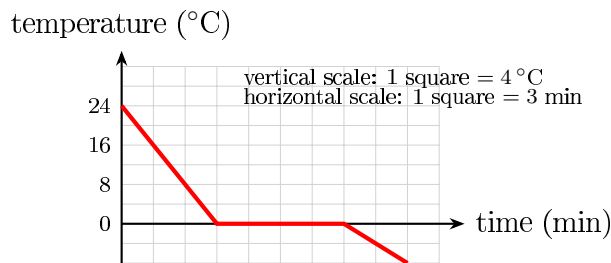
- b) What state is each substance in at room temperature (20°C)? Which state is each substance in at -100°C?
- c) What are the melting and boiling points of substance D?
- d) Could any of the substances be water?
- e) Complete the figure with a fifth substance, E, with a melting point of -80°C and a boiling point of 200°C.

Exercise 22. Luis, Carla and Javier took a piece of ice and put it in a beaker. Then they put the beaker on an electric plate and began to heat it. From time to time, they wrote down the temperature. They continued heating until the water evaporated. From their observations, they drew the following change of state graph:



When he saw the graph, Luis said that they must have done the experiment wrong because the horizontal sections indicated that the heating had stopped and that the electric plate was not working properly. His classmates told him that the plate worked perfectly and explained the reason for the horizontal sections. What did his classmates tell him?

Exercise 23. A class did an experiment about changes of state. **Group A** filled a beaker with a liquid substance and measured its temperature. Then they put the beaker in the freezer and measured the temperature every three minutes. They drew the following graph:



- What was the initial temperature of the liquid?
- What is the solidification temperature of the substance? How long did the solidification process last?
- What is the physical state of the substance after three minutes in the freezer? And after half an hour?
- How long does the substance take to freeze completely?

Part 3 — The gas laws

Exercise 24. In recipes that use pressure cookers, the cooking time is indicated, as well as the time from when steam begins to come out of the safety valve. Answer the following questions:

- a) Why is food cooked faster in pressure cookers?
- b) Do you think that is the only purpose of safety valves?
- c) What could happen if the safety valve in a pressure cooker is clogged (obstruida)?

! Important — do this in *every* gas-law exercise

In **every** gas-law exercise below, explain what is happening inside the container using the **kinetic-molecular theory**: how the particles move and how they hit the walls of the container.

Exercise 25. (Gay-Lussac) A gas is enclosed in a constant-volume container at 303 K and 2 atm. Calculate the temperature it must be heated to so that its pressure becomes 4 atm. *Sol.:* $T = 606 \text{ K} = 333^\circ \text{C}$.

Exercise 26. (Gay-Lussac) Air is cooled in a container of constant volume from 333 K to 293 K. If the initial pressure is 3 atm, what is the final pressure? *Sol.:* $P = 2.64 \text{ atm}$.

Exercise 27. (Gay-Lussac) A sample of gas occupies a volume of 44.8 litres at a temperature of 25°C and a pressure of one atmosphere. What will its pressure be at a temperature of 34°C , maintaining constant volume? *Sol.:* $P = 1.03 \text{ atm}$.

Exercise 28. (Boyle) A container with a capacity of 25 L holds a gas at a pressure of 7.5 atm. Calculate the new pressure of the gas if it is compressed to a volume of 10 L without changing the temperature. *Sol.:* $P = 18.75 \text{ atm}$.

Exercise 29. (Boyle) In a container of 5 L, oxygen gas is introduced at a pressure of 4 atm. What pressure will it exert if we double the volume without varying the temperature? *Sol.:* $P = 2 \text{ atm}$.

Exercise 30. (Boyle) When compressing a gas contained in a piston, its pressure goes from 2.3 atm to 8.5 atm. If the final volume is 2 L, what was the initial volume if the temperature remained constant? *Sol.:* $V = 7.39 \text{ L}$.

Exercise 31. (Charles) One litre of a gas is heated at constant pressure from 291 K to 331 K. What final volume will the gas occupy? *Sol.:* $V = 1.14 \text{ L}$.

Exercise 32. (Charles) A container that can vary its volume holds 12 L of a gas at 185°C . What volume will it reach if we lower the temperature to 43°C , maintaining constant pressure? *Sol.:* $V = 8.28 \text{ L}$.

Exercise 33. (Charles) A certain quantity of gas, which occupies a volume of 1 L at a temperature of 100°C , is heated up to 150°C , keeping constant pressure. What volume will it occupy under these conditions? *Sol.:* $V = 1.13 \text{ L}$.

Exercise 34. True or false? Decide whether each statement is true or false and **justify every answer using the kinetic-molecular theory** (how the particles move and how they hit the walls of the container).

- a) In a closed rigid bottle, heating the gas increases the pressure, because the particles move faster and hit the walls more often and harder.
- b) If we compress a gas to half its volume at constant temperature, its pressure also becomes half.
- c) The pressure of a gas is produced by the collisions of its particles against the walls of the container.
- d) If we cool a gas kept at constant volume, its pressure rises because the particles get closer together.
- e) The particles of a gas stop moving when the temperature reaches 0°C .
- f) When the volume of a container increases at constant temperature, the particles travel further between hits, so they strike the walls less often and the pressure goes down.

UNIT 3

Pure substances and mixtures

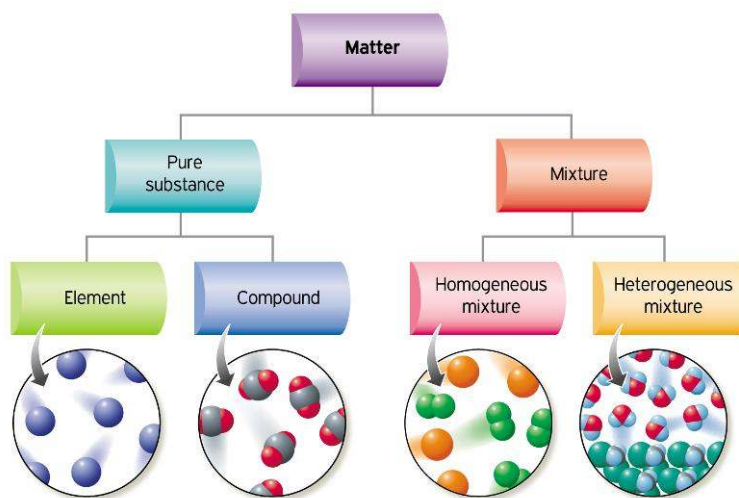
“ Is bottled water “pure”?”

In everyday life, people say pure water when they talk about drinking water or bottled water. But this water is not only water. It has H_2O molecules, but it also has particles of salts like calcium, magnesium or sodium. The composition of bottled water can vary, even if the water looks the same.

For chemists, pure water means something very strict: a liquid made only of H_2O particles, with nothing else mixed in. This kind of water is made by special purification methods, like distillation, and it is used in laboratories when truly pure substances are needed.

3.1 Classification of matter

Chemists divide matter into two main groups: **pure substances** and **mixtures**. Pure substances can be divided into two groups: **elements** and **compounds**. Mixtures can also be divided into two groups: **homogeneous** and **heterogeneous** mixtures. **Colloids** are a special type of heterogeneous mixture that we will study separately.



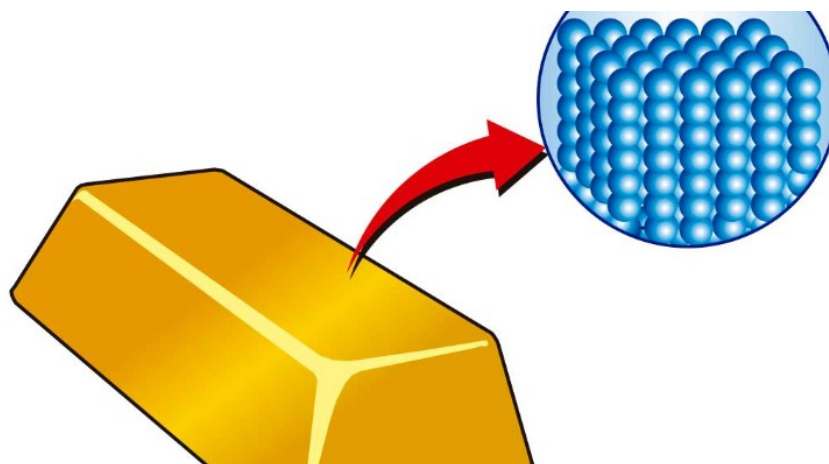
Pure substance

A **pure substance** is a type of matter made of particles that can all be the same (elements) or combinations of atoms in a fixed proportion (compounds). All pure substances have one thing in common: their composition is **fixed** and cannot change.

Elements

There are only **118 elements**, and they are all included in the **periodic table**. Knowing if a substance is an element is easy: just look at the periodic table. If it appears there, it is

an element; if it doesn't, then it is not. Examples: oxygen (O_2), gold (Au), iron (Fe).

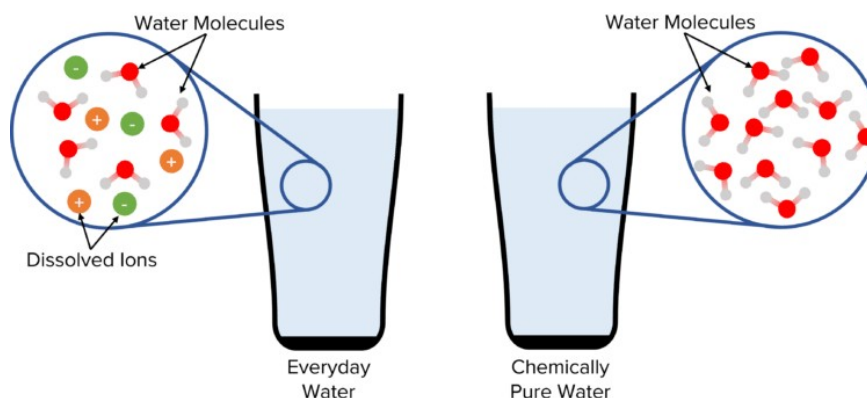


Compounds

A **compound** is a substance made of two or more elements combined in a **fixed proportion**. It is easier to understand with an example. Think about water (H_2O): in water, two hydrogen atoms **always** join with one oxygen atom. This never changes.

- If you add another oxygen atom (H_2O_2), you get a *different compound*: hydrogen peroxide (agua oxigenada).
- If you add salt to water, you now have *two substances*, not one. This is called a **mixture**. It is not a pure substance anymore, even if it looks like it.

Pure substances have properties that are always the same for them: **density**, **boiling point**, **melting point**, etc. This is very useful: we can use these properties to identify them.



3.2 Mixtures

Mixture

A **mixture** is matter made of two or more different substances. The different substances that form a mixture are called **components**. The composition of a mixture is **not fixed**: we can mix the components in any proportion.

Most everyday materials are mixtures. We divide them into two types:

Heterogeneous mixtures

You **can** see the different components of the mixture. Examples: salad, granite, water with sand, pizza.

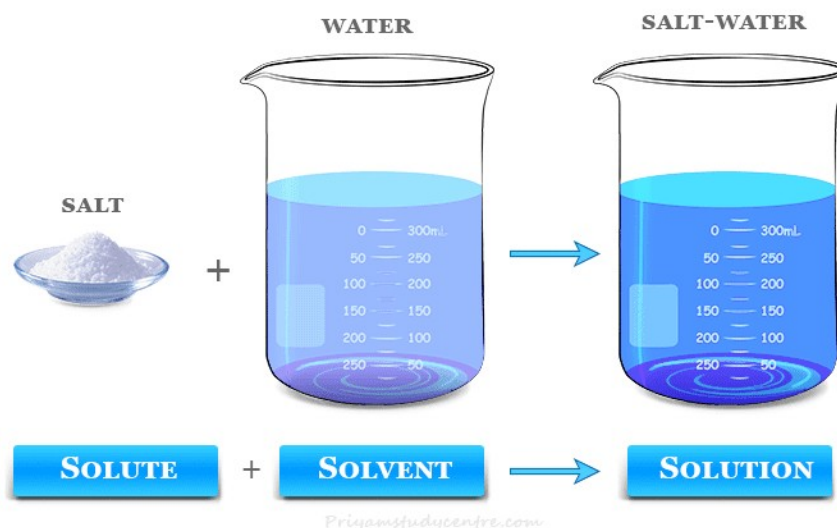
Examples of Heterogeneous Mixtures



Homogeneous mixtures (solutions)

You **cannot** distinguish the different components, because the dissolved particles are so small that they cannot be seen. Examples: salt water, air, steel.

Homogeneous mixtures are also called **solutions**. The biggest component of a solution is called the **solvent**. The smallest is called the **solute**. The physical state of the solution is the same as the state of the solvent.



Colloids

Among heterogeneous mixtures there is a special type: **colloids**. At first glance, the components cannot be seen, but if we look with a microscope, we can clearly see the particles. The smaller component of a colloid is called the **dispersed phase**, and the larger component is called the **dispersion medium**.

Depending on the physical state of the two components, colloids receive special names, such as *emulsion*, *foam*, *gel* or *aerosol*.

Some examples of colloids



Milk. It looks like a white homogeneous liquid, but under the microscope we can see tiny drops of fat spread through the water. It is an **emulsion** (one liquid dispersed in another).

Dispersed phase: fat • *Dispersion medium:* water.



Mayonnaise. It is oil with tiny drops of lemon juice inside. Oil and lemon juice are *immiscible*, so they tend to separate; to keep them together we add a stabiliser called an **emulsifier** (in mayonnaise, the egg yolk). It is an **emulsion**.

Dispersed phase: lemon juice • *Dispersion medium:* oil.



Meringue. It is made by beating egg whites with sugar. The beating traps bubbles of air in the liquid, forming a **foam**. If the meringue is baked, the egg white sets and the air stays trapped: a solid foam.

Dispersed phase: air • *Dispersion medium:* egg white.



Jelly. Fruit juice is mixed with gelatine dissolved in water. When it sets, the gelatine forms a solid full of holes and the juice is trapped inside: a liquid dispersed in a solid, which is called a **gel**.

Dispersed phase: fruit juice • *Dispersion medium:* gelatine (solid).



Smoke. It is made of very small solid particles (ash and cinders) floating in a gas. It is produced when combustion is not complete. A solid dispersed in a gas is an **aerosol**.

Dispersed phase: solid particles • *Dispersion medium:* air.



Fog. Very small drops of water float in the air near the ground, formed when water vapour cools down and condenses. A liquid dispersed in a gas is also an **aerosol**.

Dispersed phase: water • *Dispersion medium:* air.

3.3 Separation techniques

- **Elements** cannot be separated, because there is nothing simpler than an element.
- **Compounds** are difficult to separate: they can only be separated by **chemical reactions**. For example, water (H_2O) can be separated into hydrogen and oxygen by passing an electric current through it, in a process called **electrolysis**.
- **Mixtures** can be easily separated using **physical processes** (no chemical reactions needed). We will study five methods.

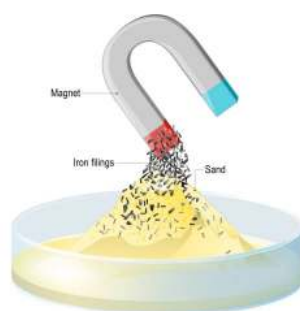


Magnetic separation

Physical principle: some materials are attracted to a magnet (like iron, nickel or cobalt), while others are not.

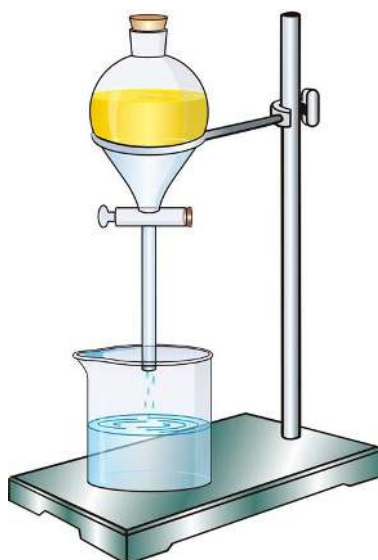
Example: separating iron filings from sand.

Separating iron filings from sand



Decantation

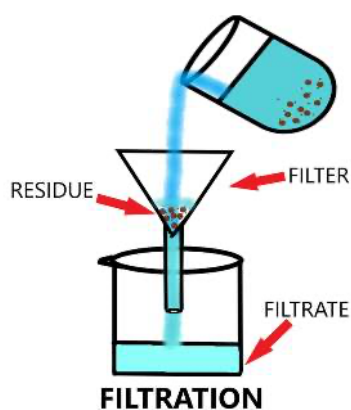
Physical principle: the difference in **density** between two liquids that do not mix. Using a separating funnel, the denser liquid (water) goes down, and the less dense one (oil) stays on top.



Filtration

Physical principle: the difference in **particle size**. Using a filter, the liquid passes through and the solid stays in the filter.

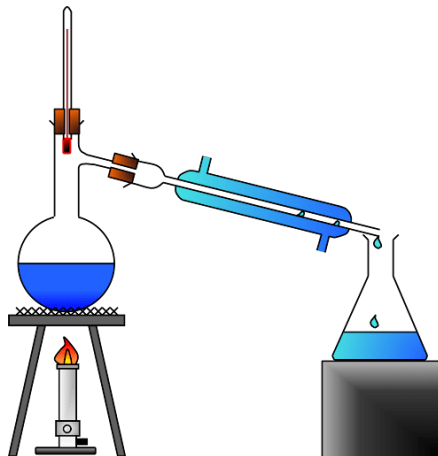
Example: separating sand from water; a colander with pasta.



Distillation

Physical principle: the difference in **boiling point**. Each substance has a specific boiling point; the component with the lower boiling point vaporizes first, and its vapour is then condensed back to liquid in a separate container.

Example: separating alcohol (boiling point 78°C) from water (100°C).



Crystallization

Physical principle: it is based on **solubility**. A dissolved substance forms solid crystals when the solution becomes supersaturated, either by cooling the solution or by evaporating the solvent.

Example: obtaining salt from sea water in salt flats (salinas).



3.4 Concentration of solutions

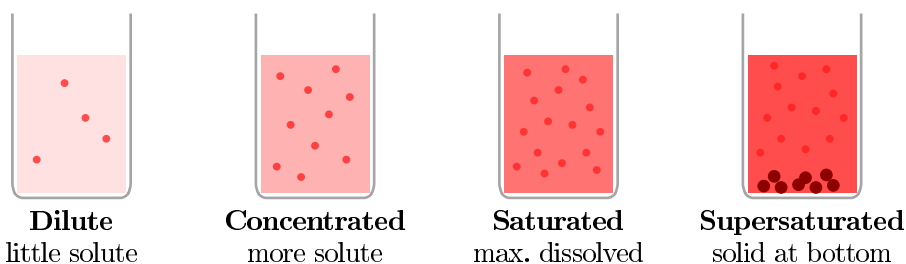
Concentration

The **concentration** of a solution indicates the proportion of solute it contains in relation to the amount of solution or solvent.

Depending on how much solute there is, we talk about:

- A **dilute solution**: little solute and a lot of solvent.
- A **concentrated solution**: a lot of solute and less solvent.
- A **saturated solution**: it has the maximum amount of solute. No more solute can dissolve — extra solute stays solid at the bottom.

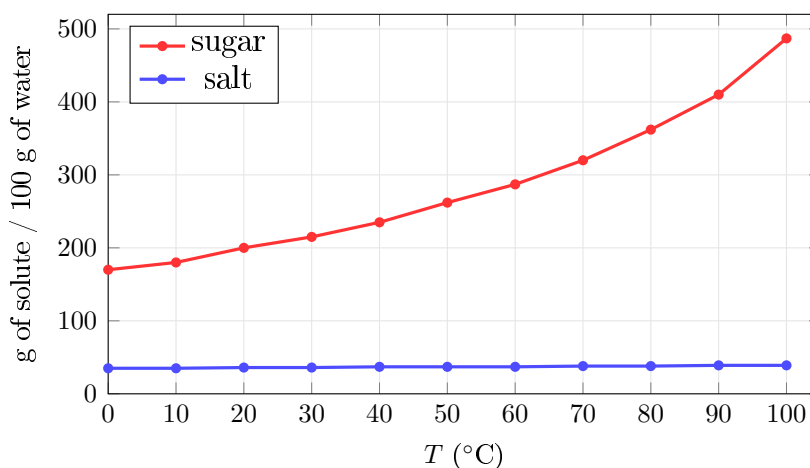
- A **supersaturated solution**: it has more solute than a saturated one. It is unstable, and the solute can crystallize easily.



Solubility

Solubility is the maximum amount of solute that can dissolve in a solvent. It is measured in grams of solute per 100 g of solvent, at a given temperature.

Why “per 100 g”? To compare substances easily. Why “at a given temperature”? Because solubility changes with temperature — usually, hot water can dissolve more solid solute than cold water.



Using solubility

The solubility of potassium nitrate (KNO_3) in water is 32 g per 100 g of water at 20 °C. How much KNO_3 can we dissolve in 250 g of water at that temperature?

$$\frac{32 \text{ g}}{100 \text{ g water}} = \frac{x}{250 \text{ g water}} \implies x = \frac{32 \cdot 250}{100} = 80 \text{ g}$$

So 250 g of water can dissolve 80 g of KNO_3 . If we add more, the extra solid will not dissolve: the solution is then **saturated**.

3.5 Concentration of a solution: the maths

We are going to study two different ways to express the concentration of a solution mathematically.

Percentage by mass

Percentage by mass means the grams of solute in every 100 g of **solution**.

$$\% \text{ by mass} = \frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \cdot 100$$

Meaning example: a solution of salt and water with a 20 % concentration by mass means that in 100 g of solution there are 20 g of salt.

! Be careful!

The mass of the **solution** is the mass of the solute **plus** the mass of the solvent: $m_{\text{solution}} = m_{\text{solute}} + m_{\text{solvent}}$. A very common mistake is to divide by the mass of the solvent only.

Mass concentration (mass per volume)

The mass concentration is the mass of solute in grams found in a given volume of solution, usually one litre.

$$\text{Mass concentration} = \frac{\text{mass of solute (g)}}{\text{volume of solution (L)}}$$

Meaning example: a solution of sugar and water with a concentration of 50 g/L means that there are 50 g of sugar in every litre of solution.

Percentage by mass

24 g of salt are dissolved in 410 g of water. Calculate the concentration of the prepared solution as a percentage by mass.

Mass of solute = 24 g

Mass of solvent = 410 g

Mass of solution = 24 + 410 = 434 g

$$\% \text{ by mass} = \frac{24}{434} \cdot 100 = 5.5 \%$$

This means: in every 100 g of this solution, there are 5.5 g of salt.

Mass concentration

15 g of sugar are dissolved in water to make a solution with a volume of 250 mL. Calculate the mass concentration of the prepared solution.

Mass of solute = 15 g

Volume of solution = 250 mL = 0.25 L

$$\text{Mass concentration} = \frac{15 \text{ g}}{0.25 \text{ L}} = 60 \text{ g/L}$$

3.6 Glossary

Glossary — 10 key terms

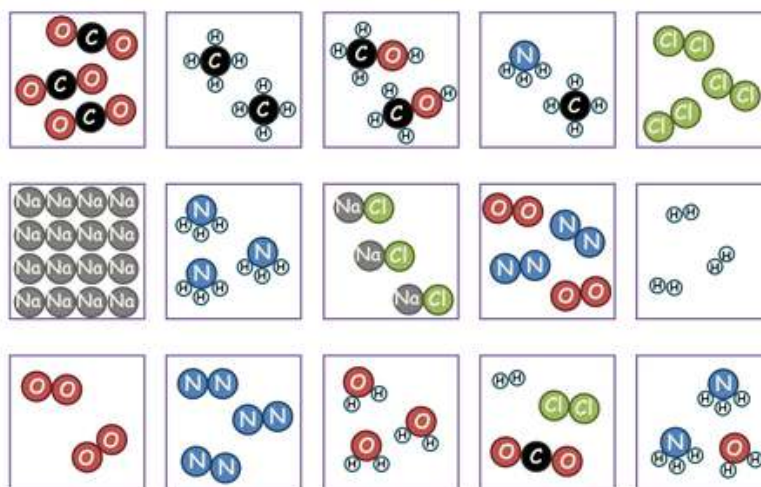
1. **Pure substance** — matter with a fixed composition that is always the same; it can be an element or a compound.
2. **Element** — a pure substance made of only one type of atom; it cannot be broken into anything simpler. Example: gold (Au).
3. **Compound** — a pure substance made of two or more elements joined in a fixed proportion. Example: water (H₂O).
4. **Mixture** — matter made of two or more substances whose proportions can change; it can be separated by physical methods.
5. **Homogeneous mixture (solution)** — a mixture that looks the same everywhere; the main component is the *solvent* and the minor one is the *solute*. Example: salt water.
6. **Heterogeneous mixture** — a mixture in which the different components can be seen. Example: sand and water.
7. **Colloid** — a mixture that looks uniform, but whose tiny particles can be seen under a microscope. Example: milk.
8. **Concentration** — the amount of solute compared with the amount of solution (or of solvent).
9. **Solubility** — the maximum amount of solute that can dissolve in a solvent at a given temperature (usually in grams per 100 g of solvent).
10. **Saturated solution** — a solution that already holds the maximum amount of solute it can dissolve; any extra solute stays solid.

3.7 Exercises

Exercise 1. Define mixture, explain how mixtures are classified, and which criteria are used for this classification.

Exercise 2. Decide if the following mixtures are homogeneous or heterogeneous: salad, freshly squeezed orange juice, mineral water and granite.

Exercise 3. Write *element*, *compound* or *mixture* below each image:



Exercise 4. Fill in the blanks using the following words: *pure substance* – *element* – *compound* – *fixed proportions* – *colloid* – *homogeneous mixture* – *heterogeneous mixture*.

A _____ can be an _____ or a _____, and its composition is always the same because the substances are combined in _____.

A _____ has the same appearance everywhere, while a _____ does not; a _____ looks uniform, but if we observe it under a microscope, we can see that it is not completely uniform.

Exercise 5. Look for information that allows you to know the composition of air, natural gas, bronze, brass and steel. In each case, indicate which is the solvent and which one or more are the solutes.

Example: Air: 78 % nitrogen, 21 % oxygen, a little less than 1 % water vapour, and traces of other gases such as argon. The solvent is nitrogen, and the solutes are the other substances.

Exercise 6. How is it possible that there are millions of compounds if there are fewer than one hundred different elements in nature?

Exercise 7. Indicate the methods to separate the components of the following mixtures. Draw a diagram of the process:

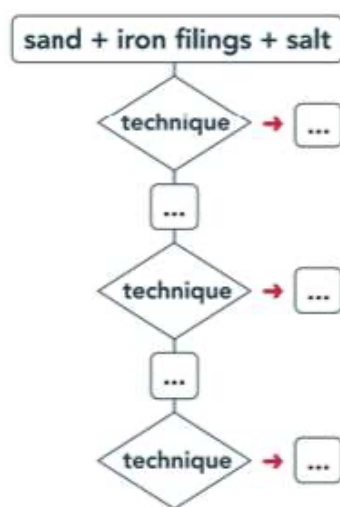
- Salt and water.
- Sawdust (serrín) and iron filings.
- Alcohol and water.
- Water and oil.
- Water, alcohol, and pieces of lead. (Note: water and alcohol are miscible.)
- Sand, water, and salt.

Exercise 8. Use the information below to answer the questions.

Data: both sugar and salt dissolve in water. Salt does not dissolve in alcohol, but sugar does.

- You have a mixture of salt and sand. How can you separate them? Write the steps, using water and alcohol if needed.
- You have a mixture of salt and sugar. How can you separate them? Write the steps, using water and alcohol if needed.

Exercise 9. You want to separate the components of a mixture composed of sand, iron filings and salt. Copy the separation diagram in your notebook:



Then, put the following terms in the corresponding place on the diagram: *salt + water*; *salt*; *sand*; *with a magnet*; *by evaporation*; *sand + salt*; *iron filings*; *add water and filter*.

Exercise 10. Read the text and answer the questions.

Petroleum is a mixture of many substances that can be separated by distillation. Some of its main substances are gasoline, kerosene, and diesel. **Gasoline** is a light fuel used in cars; it evaporates easily. **Kerosene** is slightly heavier and is used mainly in airplanes. **Diesel** is heavier than gasoline and is used in trucks and some cars. Unlike gasoline, diesel produces more pollutants when burned. By taking advantage of the different boiling points of these substances, it is possible to separate them from petroleum.

Boiling points: gasoline 100 °C; kerosene 200 °C; diesel 300 °C.

Imagine you have a sample of petroleum and you want to separate gasoline, kerosene and diesel. Using the information about boiling points, design a simple experiment:

- Decide which substance will evaporate first, second, and last.
- Choose the temperatures and the order in which you would heat the petroleum to collect each substance.

Exercise 11. Interpret the following results, obtained when calculating the concentration of several solutions:

- Sugar dissolved in water at 17 % by mass.
- Sodium hydroxide (NaOH) dissolved in water at a concentration of 30 g/L.

- c) A drink has an ethyl alcohol concentration of 36 % by volume.
- d) In a cough syrup, the medicine concentration is 0.3 mg/mL.

Exercise 12. To make a silver ring, a jeweller uses 15.73 g of pure silver and 1.27 g of copper. Calculate the mass percentage of copper in the alloy.

Exercise 13. In an experiment, 30 g of salt are dissolved in water to obtain a 150 g solution.

- a) Calculate its concentration as a mass percentage.
- b) How much salt is there in 85 g of solution?

Exercise 14. Calculate the mass of acetic acid present in 10 g of vinegar (solution) if the label shows a mass percentage of 6 % of acetic acid (solute).

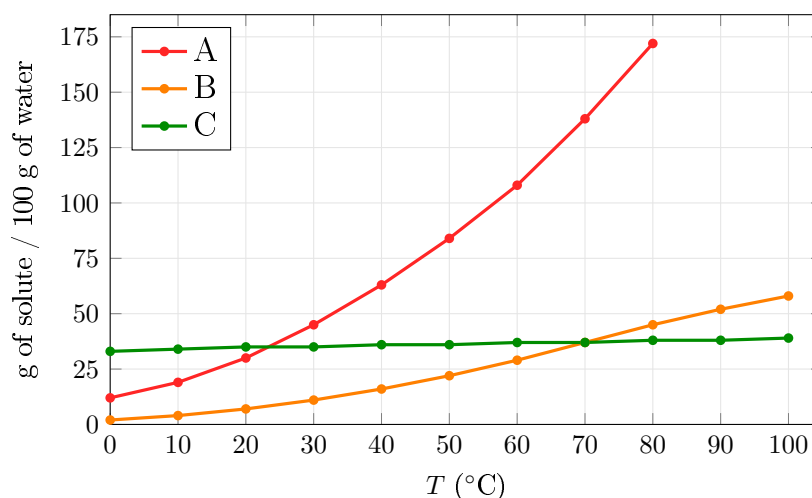
Exercise 15. A solution has been prepared by dissolving 60 g of sugar in 840 g of water, resulting in a final volume of 850 mL.

- a) Identify the solute, the solvent and the solution.
- b) What is the mass percentage of the solution?
- c) What is the concentration of the solution in grams per litre (g/L)?

Exercise 16. Calculate the mass percentage and the concentration in g/L of a solution prepared with 12 g of sugar and 300 cm³ of water. (Remember: the density of water is 1 g/cm³, and you can suppose the volume of the solution is the same as the volume of the water.)

Exercise 17. Calculate the concentration in g/L of a solution that contains 7 g of a pure substance in half a litre of solution.

Exercise 18. The following graph shows the solubility curves of three substances (A, B and C) according to the temperature of the water:



- a) Which substance is the most soluble in water at 20 °C?
- b) Which substance is the most soluble in water at 60 °C?

- c) Which substance has a solubility that is hardly affected by the temperature of the water?
- d) At what water temperature are substances B and C equally soluble?
- e) One student described the graph as follows: “The graph clearly shows substance A is the most soluble, then B, and then C is a little less soluble.” Do you think what the student said is true? Can you describe what the graph shows in a more precise way?

Exercise 19. The solubility of common salt (NaCl) is about 36 g per 100 g of water at 20 °C.

- a) How much salt can be dissolved in 50 g of water at 20 °C?
- b) If we add 25 g of salt to 50 g of water at that temperature, will all the salt dissolve? If not, how much salt stays solid at the bottom?

Exercise 20. Sugar is a substance that is highly soluble in water. Concentrated sugar solutions, called syrups, are used to prepare some desserts.

- a) Daniel, an amateur cook, made a syrup by dissolving 75 g of sugar in 100 g of water. What concentration, expressed as a percentage by mass, will the prepared syrup have?
- b) A recipe requires 100 g of a 30 % syrup by mass. What mass of sugar and water must be mixed to make this syrup?

Exercise 21. Some fish or shellfish recipes recommend cooking the food in seawater. Seawater can be unavailable or unsanitary, so it is best to prepare a brine (salmuera) that has a salt concentration similar to that of seawater.

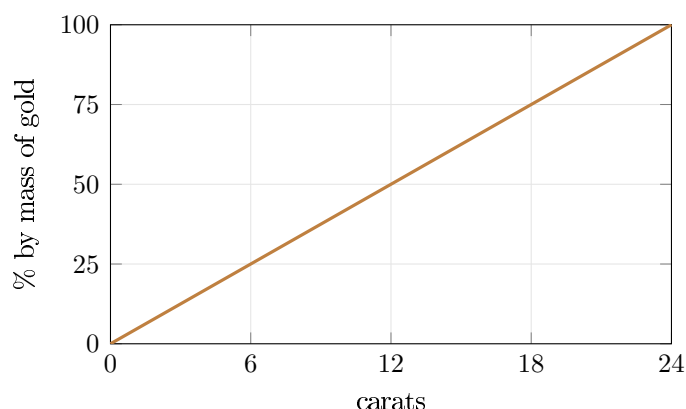
The concentration of salt in seawater is approximately 2.5 % by mass. Calculate the mass of salt and the mass of water that a cook must mix to make 2 kg of brine of 2.5 % concentration by mass.

Exercise 22. In the laboratory, four solutions of sugar water of different concentrations were prepared. The procedure followed was to separately weigh different masses of sugar and water, and then dissolve the sugar in the water. The masses mixed together are summarised in the following table:

Solution	Mass of sugar	Mass of water
A	20 g	150 g
B	0.15 kg	4 kg
C	70 g	1 kg
D	0.5 kg	425 g

- a) Say in which solution there is more dissolved sugar.
- b) Say in which there is more solution.
- c) Say which is the most dilute solution.
- d) Say which solution is the densest.

Exercise 23. The gold used in jewellery is not pure. It has been mixed with other metals, usually silver and copper. To indicate the proportion of gold in an alloy, we can use the traditional unit of measurement, called a **carat** (quilate). Pure gold is 24-carat gold. The graph shows the equivalence between carats and percentage by mass.



- How many carats is an alloy that contains 50% gold?
- 18-carat gold is the most common type of gold used in jewellery. It is called “fine gold”. What percentage of pure gold is there in a fine gold ring?
- If we mix 21 g of gold, 6 g of silver and 3 g of copper, what percentage of gold does the alloy contain? How many carats is the resulting alloy?
- What mass of pure gold is contained in a ring that has a mass of 10 g and is made of 18-carat gold?
- Why is it impossible for a ring to be 25-carat gold?

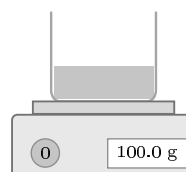
Exercise 24. One of the most important salt mines in Spain is in Remolinos (Zaragoza). The rock salt from this mine is used mainly on roads, to stop ice from forming on the coldest days. Rock salt is a mixture: it is mostly **sodium chloride** (common salt) together with **insoluble sediments**.

To find out the percentage of salt by mass in the rock, a class was divided into six groups. Each group followed the same procedure:

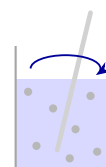
- They ground the rock salt into a fine powder with a pestle and mortar.



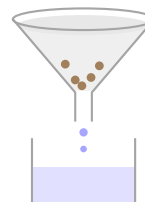
- They put a beaker on a balance, set it to zero (tared it) and weighed 100 g of powdered rock.



- They added a little more than half a litre of water and stirred well.



4. They filtered the mixture, and then dried and weighed the solid left in the filter paper.



The mass of dry solid that each group recorded is shown in the table:

Group	A	B	C	D	E	F
Solid left (g)	13	19	9	15	11	16

- Why did they add water to the powder and then filter it?
- What was the solid left in the filter paper? What was dissolved in the filtered water?
- Explain how you could recover the dissolved substance from the filtered water.
- Why do you think the groups got different results?
- If every group worked with the same care and their results were exactly correct, which group's sample contained the most salt?
- What percentage of salt by mass did each sample contain?
- Calculate the average (mean) percentage of salt in the rock salt.